

Detecting Poisoned Frames in Multi-Camera Tracking

10-Week Research Schedule

North Dakota State	Dr. Armstrong
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Student Researcher	Christine
Student Researcher	Floyd
Program	P.R.I.M.E. Summer 2026

Program Overview

This 10-week research experience investigates embedding-space poisoning in multi-camera vehicle tracking systems. The project uses 15 real intersection video clips provided by North Dakota State University as its primary dataset. Students trim and curate the footage, run baseline experiments using BoT-SORT-ReID, inject poisoned embeddings into two camera feeds, and build a cross-camera cosine distance detector to identify compromised cameras without ground truth labels.

The tracker is BoT-SORT-ReID, which extends ByteTrack with a ReID appearance embedding module. This embedding layer is the attack surface. A perturbation vector added to the embeddings from two cameras causes cross-camera identity association to silently collapse, driving down IDF1 and spiking identity switches. The detector recovers the poisoned cameras by analyzing shifts in the cosine distance distribution across camera pairs.

The goal is a conference-ready paper draft by the end of week 10, followed by an in-person presentation at North Dakota State University on the Thursday of week 10.

Weekly rhythm

Mondays are presentation days. Christine and Floyd each give a 10-minute paper presentation with Dr. Armstrong joining virtually. The format is consistent all 10 weeks: paper identity, main argument, method, key finding, limitation, and connection to the project. Annotations and slides must be committed to GitHub before 9:00 AM Monday.

Tuesdays are internal working sessions with Dr. Perry and the students only. Dr. Perry gives a brief update on the week's focus, reviews any issues from the previous week, and sets priorities for Wednesday and Thursday.

Wednesday and Thursday are independent working days. Students read, run experiments, write annotations, and prepare for Monday. Dr. Perry runs parallel research and is available by group text for quick questions and via GitHub issues for anything technical.

Fridays are virtual recaps with Dr. Armstrong and the NDSU team. Dr. Perry presents the week's progress, students summarize their work, and NDSU provides feedback and direction.

Seed papers

Four papers are assigned by Dr. Perry in weeks 1 and 2. All other sources are found independently by Christine and Floyd throughout the summer. The four seeds are: Cinà et al. (2023) Wild Patterns Reloaded (Christine, week 1); Zhang et al. (2022) ByteTrack (Floyd, week 1); Gu et al. (2019) BadNets (both, week 2); and the 2024 Neural Networks paper on adversarial attacks on multi-object trackers (both, week 2).

PHASE 1 — FOUNDATION

Week 1: Orientation and Research Landscape

Phase 1 — Foundation

Week 1 establishes the research context, introduces the GitHub workflow, and gets Christine and Floyd reading. Monday is orientation: Dr. Perry walks through the 10-week plan, shows the pre-rendered tracker demo video, and introduces the blindspot repository. Tuesday Christine receives Seed 1 (the poisoning survey) and Floyd receives Seed 2 (ByteTrack). They spend the rest of the week reading, writing their first annotation entries, and preparing their Monday presentations.

STUDENTS

Christine reads the Cinà et al. poisoning survey and prepares a 10-minute Monday presentation covering what data poisoning is, the main attack categories, and why it matters for vision systems. Floyd reads the ByteTrack paper and prepares a 10-minute Monday presentation covering how the tracker works, what MOTA and IDF1 measure, and where the pipeline could be vulnerable. Both students commit their annotation files and slides to GitHub before Monday at 11:00 AM.

DR. PERRY

Dr. Perry orients the students, runs the tracker demo video, distributes seed papers with a brief written note explaining why each was chosen, and walks through the annotation template and Monday presentation format. Dr. Perry also begins independent literature review on embedding-space attacks and detection methods.

NDSU — DR. ARMSTRONG

Dr. Armstrong joins on Monday virtually. The week 1 Monday session is orientation — no paper presentations yet. Dr. Armstrong meets Christine and Floyd, hears the project overview, and the Friday recap covers week 1 orientation and the plan for week 2.

Week 2 Threat Modeling and the ReID Attack Surface

Phase 1 — Foundation

Week 2 introduces the attack surface through a live demo and asks students to build their first threat model. On Tuesday Dr. Perry runs the BoT-SORT-ReID demo on NDSU intersection footage — showing bounding boxes and track IDs, printing an embedding vector to the terminal, demonstrating cosine similarity between matching and non-matching vehicle crops, and showing what happens when a perturbation vector is added. Students then build independent threat model diagrams of the full pipeline.

STUDENTS

Christine and Floyd both read Seeds 3 and 4 — BadNets and the adversarial MOT paper. They attend the live demo, discuss how BadNets backdoor mechanics translate to embedding-space poisoning, and independently draw a threat model diagram of the BoT-SORT-ReID pipeline showing all attack entry points. Each student also finds one additional paper independently and commits an annotation entry. Monday presentations cover Seeds 3 and 4 plus the threat model diagrams.

DR. PERRY

Dr. Perry runs the live BoT-SORT-ReID demo, walks through the cosine similarity demonstration, and facilitates the threat model discussion. Dr. Perry also begins drafting the problem formalization section for the paper and decides on the clip trimming standard for week 3.

NDSU — DR. ARMSTRONG

Dr. Armstrong joins Monday for presentations on Seeds 3 and 4 and the threat model diagrams. The Friday recap covers the demo results and the planned dataset curation approach for week 3.

PHASE 2 — EXPLORATION

Week 3 Dataset Curation and Clean Baseline

Phase 2 — Exploration

Week 3 produces the blindspot dataset and establishes the unpoisoned control baseline. Christine and Floyd watch through the 15 NDSU intersection clips, select the best 2-minute segment from each one — prioritizing scenes with multiple vehicles appearing across more than one camera view — trim the clips using ffmpeg, and commit them to the repository. Dr. Perry runs BoT-SORT-ReID on the trimmed clips and logs the clean baseline metrics.

STUDENTS

Christine reviews clips 1 through 7 and Floyd reviews clips 8 through 15. For each clip they select a 2-minute segment, trim it with ffmpeg, write a short markdown note explaining the segment choice, and commit everything to the data folder. They also find 1 to 2 papers independently on multi-camera tracking evaluation or ReID benchmarks and commit annotation entries.

DR. PERRY

Dr. Perry runs BoT-SORT-ReID on student-trimmed clips as they arrive, logs IDF1, HOTA, MOTA, and IDS for each clip, and builds the baseline metrics table in the results folder. Dr. Perry also drafts the dataset description section of the paper.

NDSU — DR. ARMSTRONG

Monday presentations cover papers on multi-camera tracking or ReID. The Friday recap shows Dr. Armstrong the trimmed clips and baseline tracker output and discusses whether any clips appear to be synchronized multi-camera footage of the same intersection.

Week 4 Embedding-Space Poisoning — Two-Camera Attack

Phase 2 — Exploration

Week 4 runs the core poisoning experiment. The EmbeddingPoisoner hook intercepts the OSNet embedding extraction step in BoT-SORT-ReID and adds a perturbation vector to the output of two designated clips, treating them as two compromised cameras. The experiment runs at three perturbation magnitudes and the results are compared against the week 3 clean baseline.

STUDENTS

Christine and Floyd integrate the EmbeddingPoisoner into the tracking loop, run experiments at epsilon values of 0.1, 0.5, and 1.0 on two designated clips, log all metrics, and build the comparison table. They also find one paper on adversarial ReID or embedding-space attacks and commit an annotation entry.

DR. PERRY

Dr. Perry guides the poison hook integration, runs parallel targeted perturbation experiments, and drafts the threat model section for the paper. Dr. Perry reviews student metric logs before the Friday recap.

NDSU — DR. ARMSTRONG

Monday presentations cover papers on adversarial tracking or ReID. The Friday recap presents the poisoning results table to Dr. Armstrong — clean baseline versus poisoned at each epsilon level.

PHASE 3 — ASSESSMENT AND BUILD

Week 5 Midpoint Assessment

Phase 3 — Assessment and Build

Week 5 is the formal halfway review. Christine and Floyd each give a 20-minute midpoint presentation on Monday covering everything from weeks 1 through 4 — literature, experimental results, and a detection hypothesis. Dr. Perry completes a written rubric assessment for each student. The Friday recap brings Dr. Armstrong into the conversation about the paper's direction and the contribution statement for the second half.

STUDENTS

Both students prepare a 20-minute presentation covering their literature findings, experimental contributions, and one concrete detection hypothesis. The annotated bibliography should be at 8 or more sources per student before this session.

DR. PERRY

Dr. Perry completes the Week 5 rubric assessment with written narrative for each student, writes an updated research direction memo, and prepares a draft paper abstract for Dr. Armstrong to review on Friday.

NDSU — DR. ARMSTRONG

Dr. Armstrong joins the extended Monday session for both midpoint presentations and gives feedback. The Friday recap is a substantive direction-setting conversation — Dr. Armstrong reviews the draft abstract and contribution statement and the group agrees on the detection approach to pursue in weeks 6 and 7.

Week 6 Cross-Camera Embedding Consistency Detector

Phase 3 — Assessment and Build

Week 6 builds the detection method. The cross-camera cosine distance detector works by collecting pairwise cosine distances between embeddings the tracker assigns to the same global identity across camera pairs. Poisoned cameras produce a systematically shifted distance distribution. The detector flags cameras whose mean distance is statistically above the group mean — without requiring any ground truth identity labels.

STUDENTS

Christine and Floyd implement the CrossCameraDetector, integrate it into the tracking loop, run it against the week 4 poisoned scenarios, and produce precision, recall, and F1 scores for poisoned clip identification at all three epsilon levels. They also generate cosine distance distribution plots showing the shift in poisoned versus clean clips.

DR. PERRY

Dr. Perry surveys the detection literature more deeply, identifies 2 to 3 prior approaches for comparison in the paper, and drafts the related work section outline.

NDSU — DR. ARMSTRONG

Monday presentations cover papers on anomaly detection or distribution shift. The Friday recap presents the detector results and distribution plots to Dr. Armstrong.

Week 7 Sustainability and Scalability Review

Phase 3 — Assessment and Build

Week 7 stress-tests the detector across all NDSU clips and boundary conditions to determine where it holds and where it breaks. Students vary the number of poisoned clips from 1 to N, test on clips not used in week 4, and document every failure mode with a mechanistic hypothesis. By the end of this week the paper's core contribution statement is agreed on by Dr. Perry and Dr. Armstrong.

STUDENTS

Christine and Floyd run the detector on all trimmed NDSU clips, vary the number of poisoned clips, compare random perturbation versus targeted perturbation, and write a scalability report documenting performance across varied conditions and failure mode analysis.

DR. PERRY

Dr. Perry runs targeted perturbation scalability tests in parallel, drafts the paper contribution statement for Friday review, and begins the paper structure outline.

NDSU — DR. ARMSTRONG

Monday presentations cover papers on robustness and generalization. The Friday recap is the contribution-locking session — Dr. Armstrong reviews the scalability results and the group agrees on the one-sentence paper contribution statement.

PHASE 4 — WRITING AND PUBLICATION

Week 8 Paper Draft — Literature Review and Introduction

Phase 4 — Writing and Publication

Writing begins in week 8. Christine and Floyd draft the literature review organized into four thematic pillars: data poisoning foundations, multi-object tracking and ReID, multi-camera tracking and MTMC benchmarks, and adversarial attacks on trackers and detection methods. The seed papers anchor each theme and the student-found papers build outward from there. Dr. Perry drafts the introduction.

STUDENTS

Christine and Floyd each draft their assigned sections of the literature review, submit drafts for peer review before Thursday, and continue finding papers to close any remaining citation gaps.

DR. PERRY

Dr. Perry drafts the paper introduction and argument structure, peer reviews both students' literature review sections with written feedback, and ensures the contribution statement is reflected consistently.

NDSU — DR. ARMSTRONG

Monday presentations cover any remaining citation gap papers. The Friday recap is a writing review session — Dr. Armstrong reads the introduction draft and gives feedback on the contribution framing.

Week 9 Methods, Results, and Contribution Framing

Phase 4 — Writing and Publication

Week 9 produces the full rough draft. Christine and Floyd write the methods and results sections covering dataset curation, the baseline experiment, the poisoning attack, the detector, and the scalability tests. Dr. Perry synthesizes everything into one integrated document, writes the abstract and conclusion, and ensures the contribution is stated consistently from introduction through discussion.

STUDENTS

Christine and Floyd draft their assigned methods and results sections, integrate everything into a single document, and participate in the full paper read-through on Friday where each person flags one unclear element and one strong element.

DR. PERRY

Dr. Perry synthesizes the full draft, writes the abstract and conclusion, and prepares a revision plan assigning section ownership for week 10.

NDSU — DR. ARMSTRONG

Monday presentations cover final citation gap papers. The Friday recap is a full paper review — Dr. Armstrong reads the complete draft and gives feedback before the final revision week.

Week 10 Revision, Venue Targeting, and In-Person Presentation

Phase 4 — Writing and Publication

Week 10 revises the paper based on Dr. Armstrong's feedback and prepares the final presentation. On Thursday the full team travels to North Dakota State University to deliver the in-person final presentation. Dr. Perry identifies 2 to 3 target publication venues and assigns post-program revision tasks so the paper has a clear submission path after the experience ends.

STUDENTS

Christine and Floyd revise their assigned sections, prepare and deliver the in-person NDSU presentation, and write a personal reflection on what they would pursue next and what they would do differently.

DR. PERRY

Dr. Perry finalizes the paper revision, identifies target venues with submission deadlines, leads the in-person presentation at NDSU, and assigns post-program revision tasks with clear ownership and deadlines.

NDSU — DR. ARMSTRONG

Monday is a final virtual check-in to confirm travel logistics and the presentation plan. Thursday the full team is in person at NDSU — the final presentation covers the complete research story from problem definition through detection results and future work. Dr. Armstrong and the NDSU team provide closing feedback and the group confirms the publication timeline.